HEKMA AI PRIVACY MICROSERVICE

RoBERTa PII / PHI Model Performance & Throughput Benchmark Report

Version: 2.0.0 | **Model:** fine-tuned RoBERTa-base (25 PII Entities) | **Evaluation Date:** August 2026 | **Target Domain:** https://pii-phi.hekma.ai

1. Executive Summary & Key Highlights

This report provides rigorous empirical benchmarking and scale extrapolations for the Hekma AI Privacy microservice. The service executes neural token classification via a fine-tuned RoBERTa model for demographic PII detection, combined with high-throughput reversible surrogate token generation, AES-256-GCM Token Vault encryption, and FHIR/JSON structured data anonymization.

Single Record Latency **19.1 ms** **Sub-20ms SLA for real-time EHR API ingress**	**Batch Throughput (8 vCPU)** **~1,100 rec/sec** **High-throughput parallel worker scaling**	**Structured FHIR Ingress** **~2,500 rec/sec** **Fasten Health JSON tree transformation**	**1 Million Records Time** **15.2 min (CPU)** **5.8 min on GPU acceleration**

2. Micro-Benchmark: Pipeline Latency Breakdown

Tests were executed on clinical note snippets averaging 150–300 characters / 35–60 words per record (containing patient names, SSNs, dates of birth, MRNs, addresses, and clinician details).

Pipeline Processing Stage	Sequential (1 Record)	Batch-64 (Per Record)	% Total Time	Technical Description
1. Tokenization & Subword Encoding	1.52 ms	0.31 ms	8.0%	Regex word boundary splitting + Byte-Pair Encoding (BPE) tensor generation
2. RoBERTa Neural Inference (CPU)	16.48 ms	6.20 ms	86.3%	Transformer self-attention layer forward pass & softmax BIO probability extraction
3. BIO Tag Merging & Offset Map	0.60 ms	0.41 ms	3.1%	Reconstruction of multi-token spans (B-PERSON + I-PERSON) to exact char offsets
4. Token Vault & Masking Engine	0.50 ms	0.38 ms	2.6%	Cryptographic surrogate token mapping, AES-256 encryption, string slicing
Total End-to-End Latency	**19.10 ms**	**7.30 ms**	**100.0%**	**Full predict + surrogate tokenization + masking ready for downstream storage**

3. Processing Time & Throughput by Data Volume (100 to 1 Million)

The following matrix outlines the projected execution time to predict and anonymize datasets ranging from 100 records to 1,000,000 records across 3 operational infrastructure tiers:

Dataset Volume	Mode 1: Single CPU Worker (Sequential / 1 Core)	Mode 2: 8 vCPU Multi-Core (Uvicorn 8 Workers, Batch-64)	Mode 3: GPU Accelerated (NVIDIA T4 / A10G TensorRT)	Recommended Processing Strategy
100 records	1.91 seconds	**~90 ms (0.09s)**	**~35 ms** (0.035s)	Synchronous REST API call (/predict or /anonymize)
1,000 records (1k)	19.10 seconds	**0.91 seconds**	**0.35 seconds**	Batch API endpoint (/predict/batch) in single payload
10,000 records (10k)	3.18 minutes (191s)	**9.12 seconds**	**3.50 seconds**	Chunked concurrent worker ingress or streaming queue
100,000 records (100k)	31.83 minutes	**1.52 minutes (91s)**	**35.0 seconds**	Async Celery / AWS SQS task pipeline with DB bulk upsert
1,000,000 records (1M)	5.30 hours	**15.20 minutes**	**5.83 minutes**	Distributed batch migration / IONOS K8s horizontal scale

4. Workload Characteristics Across Data Modalities

Data Modality / Source	Engine Mechanism	Throughput (Per Core)	1 Million Records Time
1. Pure Structured FHIR (Fasten EHR: Patients, Telecoms, Identifiers)	Schema-aware JSON tree traversal, deterministic policy lookup, token substitution without neural NLP.	**1,500 – 2,500 rec/s**	**~6.5 – 11 minutes (Single 8-core node)**
2. Mixed FHIR + Clinical Notes (DiagnosticReport, DocumentReference)	Direct schema anonymization for structured attributes + RoBERTa NLP on free-text clinical narrative.	**400 – 800 rec/s**	**~20 – 40 minutes (Single 8-core node)**
3. Unstructured Clinical Text (Doctor Progress Notes, Medical Transcripts)	Full token classification across entire document length with multi-span entity boundary alignment.	**137 rec/s (batch-64)**	**~15.2 minutes (8 parallel workers)**
4. OpenAI PHI Detector (Optional LLM gpt-4o-mini Backend)	Remote OpenAI API call. Subject to HTTP round-trip (~350ms) and OpenAI TPM tier quotas.	**~50 – 100 req/s** (50 concurrent async)	**Best for real-time / batch offline** (Switch to local for 1M bulk)

5. Production Sizing & High-Throughput Ingestion Recommendations

• **Uvicorn Worker Sizing:** Run uvicorn api.main:app --workers 8 on an 8-vCPU node (e.g. AWS c6i.2xlarge or IONOS Cloud). This unlocks ~1,100 records/second sustained throughput without GPU cost.

• **Batch Ingestion Endpoints:** For datasets > 1,000 records, invoke POST /predict/batch or POST /api/v1/anonymize/fhir/batch with batch_size=64 or 128 to achieve 2.6x higher computational efficiency over sequential single-record loops.

• **PostgreSQL Token Vault Connection Pooling:** Set DB_POOL_SIZE=20 and DB_MAX_OVERFLOW=30 to handle concurrent token registration without database I/O bottlenecks during 1M record ETL migrations.

• **Memory Footprint:** The RoBERTa model weights occupy ~490 MB in RAM per worker process. An 8-worker deployment requires ~4.5 GB of system RAM, fitting comfortably within an 8 GB or 16 GB instance.

• **Sub-50ms SLA Compliance:** Real-time doctor note ingress via EHR webhooks executes in ~19.1 ms, providing comfortable headroom beneath standard 50 ms API latency targets.